

# NanoGPT speedrun (record #40): validation loss vs wall-clock



- record #40 upstream (mean of 5)
- Muon  $\text{Ir}=0.06$
- SignSGD  $\text{Ir}=0.03$
- ▲ SignMuon  $\text{Ir}=0.03$
- ◆ MuonUSign  $\text{Ir}=0.06$
- + EF21-SignMuon  $\text{Ir}=0.06$
- ▼ MuonSign  $\text{Ir}=0.03$
- × EF21-MuonUSign  $\text{Ir}=0.06$
- EF21-MuonSign  $\text{Ir}=0.06$